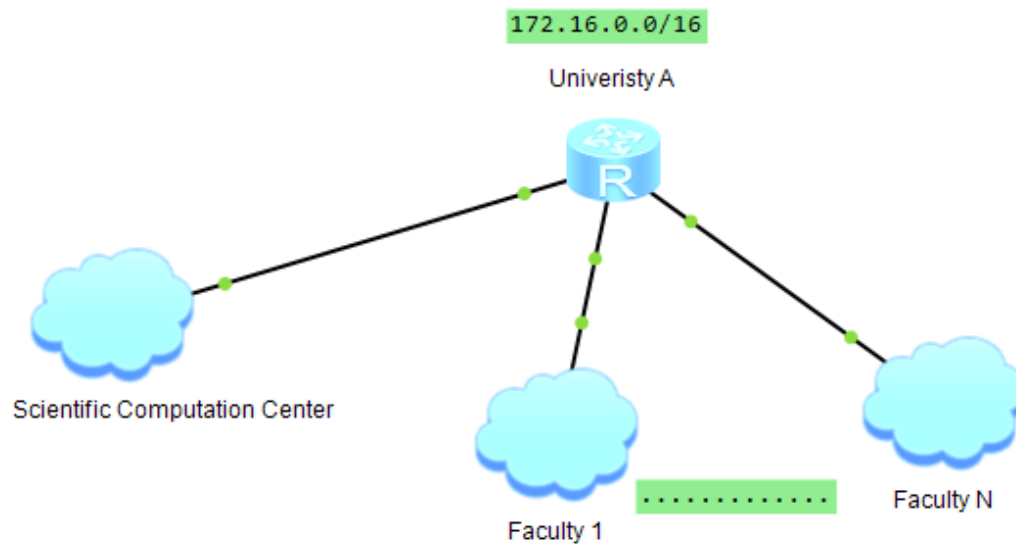


Computer Networks Course Project

University Campus Network Design

Description: Design a network for a small university campus



- **University A** has several faculties, at minimum **four**, and **one Scientific Computation Center**.
- Scientific Computation Center contains **several servers** that all connected with a **switch**.
- Each faculty is composed of several floors, at a minimum of **four**.
- Each floor is composed of several labs, at minimum **five**.
- Each lab contains at least **ten** PCs, all these PCs are connected with **switch**.
- All the switches of the labs on the **same** floor are connected with a **switch**.
- All the floors are connected by a switch that connects with **the main router** of the university.
- The **university router** connects all the faculties and the Scientific Computation Center with each other and with the world wide web.

Please, Apply the following points:

- Use main network (172.16.0.0/16) to distribute the **private IP addresses** to all sub-networks (Faculty **1**, faculty **2**, faculty **N**) and **Scientific Computation Center according to your topology**. You must calculate **network address** and **broadcast address** of each **faculty subnet** according to your topology.
- Apply VLAN configuration in your network as each lab number **1** in each floor should be assigned to VLAN **10**, Each lab number **2** in each floor should be assigned to VLAN **20** and Each lab number **3** in each floor should be assigned to VLAN **30**.
- Write three test scenarios to evaluate your network in addition to these scenario examples:
 - **Test reachability is successful in the next Scenario:**
 - Pc number 0** on the **first** floor in the **second** lab in the faculty of computers & Artificial Intelligence want to access data on the first **server** in the Scientific Computation Center.
 - ❖ Source IP address: **Pc 0** IP address.
 - ❖ Destination IP address: **Server 0** IP address.
 - Test that all labs in the same Vlan are reachable to each other or not. justify your result.
 - Test that all labs in the different Vlan are reachable to each other or not. justify your result.

Bonus: OSPF Routing Protocol (3 grades)

OSPF (Open Shortest Path First) Protocol is implemented for routing within the main university router to route data between different faculties subnets and the Scientific Computation Center subnet.

Project supervised by:

Prof. Heba M Abd EL Atty